

SOUTHEAST HAWAII BUOY 51004, USA

WMO: 992620

Lat: 17.533N

Lon: 152.255W

Elev: 0

StdP: 101.33

Time Zone: -10.00 (W10)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.5	22.0	N/A	N/A	N/A	N/A	N/A	N/A	13.4	23.1	12.0	23.3	8.1	60	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.6	28.6	N/A	28.3	N/A	27.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	11.0	10.5	DB	19.4	27.7	2.6	1.2	17.6	28.5	16.0	29.2	14.6	29.9	12.7	30.7	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.7	23.9	23.7	23.7	24.0	24.5	25.0	25.4	25.9	25.8	25.6	25.0	24.3
	DBStd	1.17	0.77	0.90	0.84	0.84	0.91	1.09	1.05	0.88	0.84	0.76	0.74	0.69
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5376	432	382	425	419	449	449	477	492	474	482	451	442
	CDD18.3	2334	173	149	167	169	191	199	219	233	224	224	201	184
	CDH23.3	14279	612	451	514	672	1013	1333	1695	2014	1913	1804	1376	882
	CDH26.7	443	0	0	0	0	8	79	97	119	84	42	13	1
Wind	WSAvg	7.6	7.3	7.6	7.5	8.0	7.1	7.4	7.6	7.7	7.0	7.4	7.9	8.3
Precipitation	PrecAvg	361	47	38	46	13	11	6	14	29	30	32	45	48
	PrecMax	753	120	159	190	78	64	30	54	203	138	96	143	298
	PrecMin	154	1	4	2	1	0	2	2	3	2	2	0	1
	PrecStd	141	38	34	45	16	13	5	10	39	35	21	36	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.4	26.1	26.2	26.3	27.5	29.1	29.0	29.0	28.8	28.2	28.0	26.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	25.8	25.6	25.7	25.8	26.9	28.6	28.5	28.4	28.0	27.7	26.9	26.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	25.5	25.2	25.3	25.7	26.5	27.9	28.0	28.0	27.7	27.2	26.5	25.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	25.2	25.1	25.1	25.4	26.1	26.4	27.0	27.4	27.1	26.8	26.2	25.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	1.7	1.7	1.6	1.7	1.6	1.8	1.7	1.6	1.5	1.4	1.4	1.5
		MCDBR	2.2	1.8	1.8	2.3	2.1	2.1	2.0	2.2	2.3	2.0	1.9	2.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Clear-Sky Solar Irradiance	taub	0.359	0.365	0.372	0.404	0.396	0.382	0.382	0.384	0.373	0.372	0.372	0.362
taud		2.606	2.588	2.569	2.443	2.486	2.519	2.543	2.546	2.566	2.609	2.605	2.621	
Ebn at Noon		930	940	939	903	898	902	903	909	924	921	910	917	
Edh at Noon		92	98	103	117	111	106	104	104	102	95	92	89	
All-Sky Solar Radiation	RadAvg	4.78	5.29	5.93	6.55	6.85	6.82	6.88	6.78	6.53	5.57	4.67	4.43	
	RadStd	0.27	0.34	0.50	0.40	0.28	0.29	0.23	0.25	0.26	0.21	0.34	0.33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page